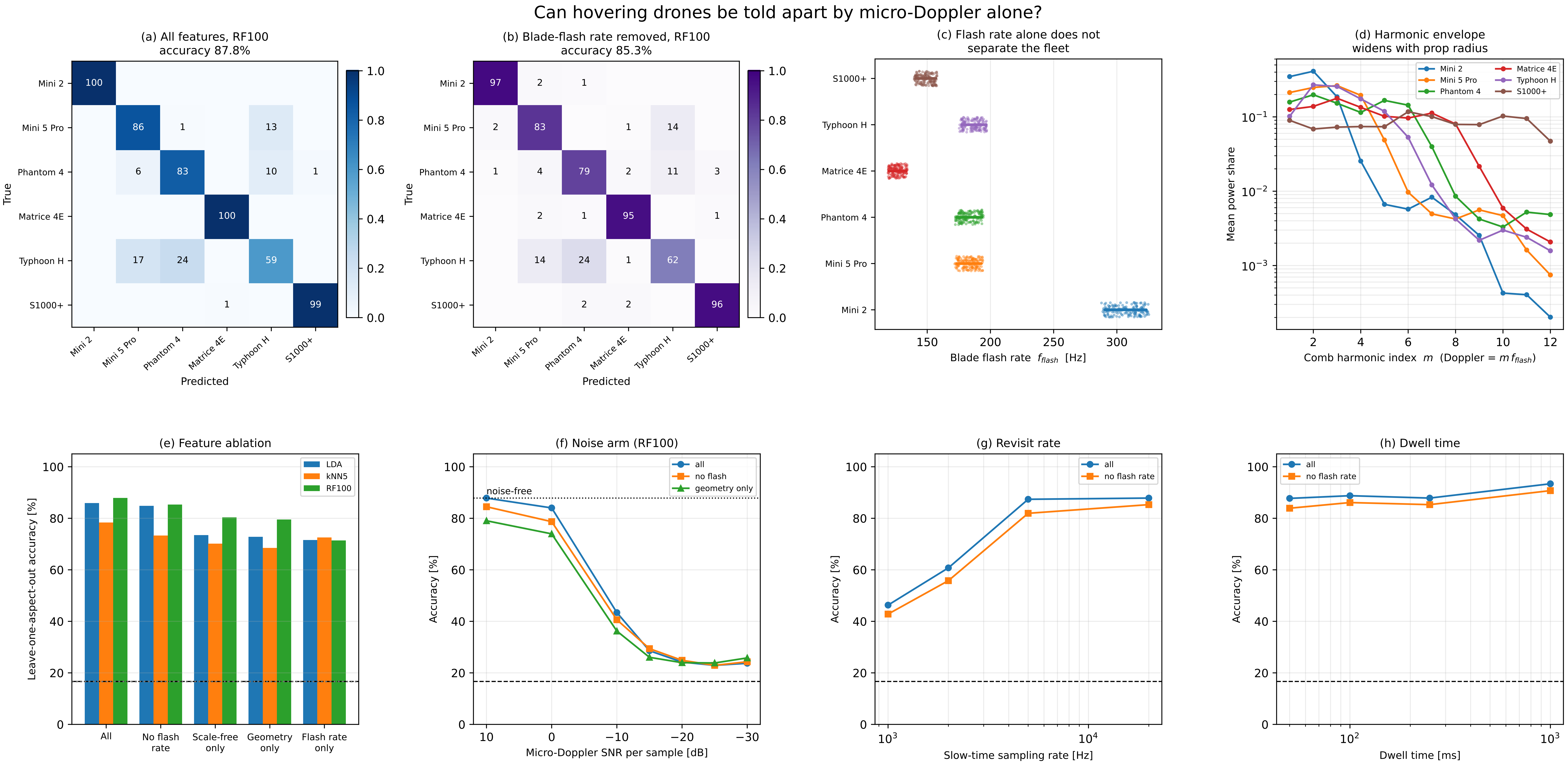

Shooting-and-bouncing-rays micro-Doppler at 20 kHz slow-time rate and 250 ms dwell: 864 trials over 18 aspects and 6 airframes. Hover rpm is redrawn +/-6% every trial and the per-rotor spread is redrawn too, so the blade flash rate is not a class label -- panel (c) shows how far the rate distributions overlap. Cross-validation holds out whole aspects, so every test geometry is unseen. Label-permutation control lands at 16.6% against a 16.7% chance line, so the aspect-held-out split does not leak. The simulation carries no noise and no clutter, so panels (a), (b), (e) are an upper bound and panel (f) is the honest curve.